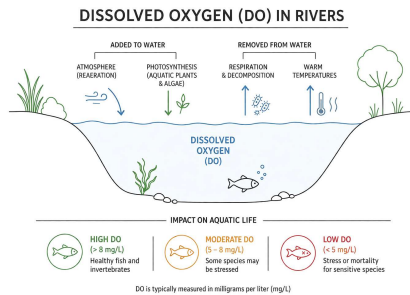


Forecasting in the Logan River

Using ARIMA modeling to predict Dissolved Oxygen Levels

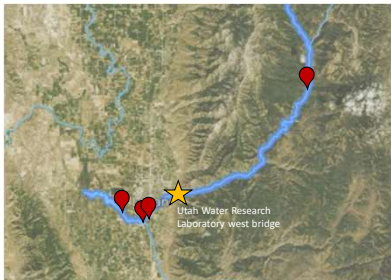
What is Dissolved Oxygen?



Goal: Explore different predictive models to best predict dissolved oxygen levels in the Logan River.

Data

- Data comes from an open-source platform, HydroServer, from the Logan River Observatory
- Multiple sensors are located along the Logan River, below are the sensors that measure dissolved oxygen levels in the Logan River, starred is sensor used in measurements



Modeling

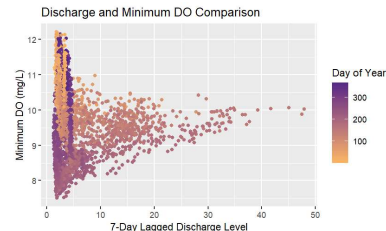
We can model the trend in the minimum level of dissolved oxygen for every day over the seasons using an ARIMA model

- Data from the years 2014-01-01 to 2024-02-14 were used to fit the model
- Data from 2024-02-15 to 2025-12-04 were then used to test the model's accuracy
- Two models were fit:
 - Model 1 was fitted using the overall trend of the dissolved oxygen levels
 - Model 2 was fitted using the overall trend and the previous week's discharge levels

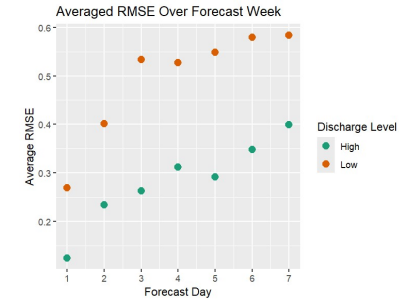
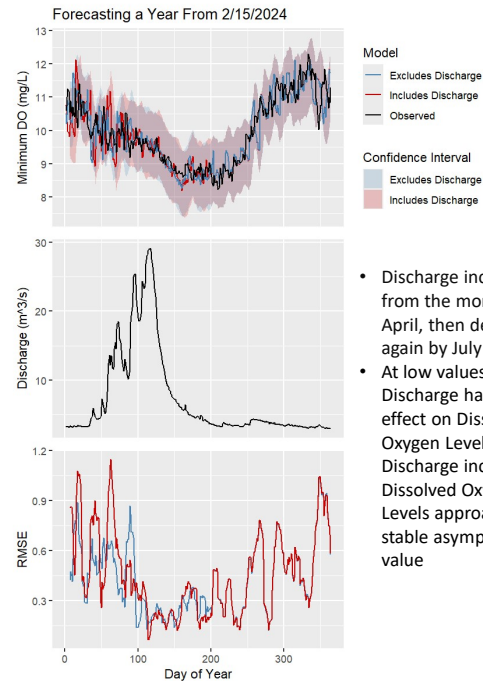
$$Y_t = \phi_0 + \sum_p \phi_p Y_{t-p} + \beta_1 Y_{t-365} + \sum_q \theta_q \epsilon_{t-q} + \gamma X_{t-7} + \epsilon_t$$

Y_t : Dissolved Oxygen at time t
 ϕ_0 : Baseline Dissolved Oxygen level
 $\sum_p \phi_p Y_{t-p}$: Dependence on previous p day(s)
 $\beta_1 Y_{t-365}$: Dependence on same day of previous year
 $\sum_q \theta_q \epsilon_{t-q}$: Dependence on previous q day(s) error term
 γX_{t-7} : Effect of river Discharge (only used for model 2)
 ϵ_t : Random noise

- Forecasts were made weekly, updating the model with new data before each next-week prediction
- Start of the prediction year, there is a difference between the two model's predictions
- At the end of the prediction year, the two models converge their predictions



- At the start of the prediction year, when discharge is low, RMSE of the model with discharge varies a lot and can be very high compared to the model without
- When discharge is high, RMSE of the model with discharge decreases drastically



- Discharge level (m^3/s) is split to be high when greater than 10 and low for lower than 10
- RMSE is averaged for each day of the week for the model including discharge
- RMSE increases as week progresses, less accurate

Results

- Overall, both models predict very well, as the observed values are within each confidence interval
- When discharge is high, the accuracy of predicting dissolved oxygen levels increases for both models
- For long trends of low discharge levels, both models converge to the same prediction

Conclusion

- Predicting dissolved oxygen levels is important for maintaining the health of a river's ecosystem.
- The previous weeks discharge levels are only useful while discharge is high and is basically irrelevant to the predictions when low.

Future goals:

- Using the same model to predict at different sites along the Logan River
- Fit a model using air temperature and precipitation, as this is easily forecasted